

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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ASSESSMENT TOOL 1 – ANSWER SHEET

Name: Shanmuganathan S R S Register No: 192511347

Course: Artificial Intelligence (CSA17) CO4

Note: The question paper specifies the dataset fields but does not provide the actual dataset or measured model outputs. Therefore, the numerical evaluation/Q-table values below are a clearly stated **illustrative worked example**; actual experimental values should be replaced when the dataset is available.

Task 1: Data Preparation & Inductive Learning

(a) Inductive learning approach

Use supervised inductive learning because labelled historical customer records are available. The model learns a general rule from past examples and predicts the class of a new customer. The target variable is **Loan Default** (Default/No Default). Key features are: **credit score** because it summarizes repayment risk; **existing debts** because high debt increases financial burden; **repayment history** because previous missed payments are a strong indicator of future default; and **income/loan amount** because they indicate repayment capacity and debt burden.

(b) Class imbalance

Since defaulters are much fewer than non-defaulters, accuracy alone can be misleading. I would use **SMOTE on the training set** to create synthetic minority examples, combined with stratified splitting and class-weighted evaluation. SMOTE improves minority-class recall without discarding many genuine non-default examples. The test set must remain untouched and retain the original distribution.

(c) 70:30 split and preprocessing

Use a **70% training / 30% testing** split with stratification. The larger training portion provides enough examples to learn rare default patterns, while 30% gives a reliable independent evaluation. Numerical variables can be standardized using z-score normalization. Categorical variables such as employment type are one-hot encoded. Scaling prevents large-unit variables from dominating statistical models; encoding converts categories into numerical inputs. For a Decision Tree, scaling is not essential, but the same preprocessing pipeline can be maintained for fair model comparison.

Task 2: Decision Tree for Loan Default Prediction

(a) Tree and root-node selection

A Decision Tree recursively chooses the feature that produces the largest impurity reduction. For classification, Gini impurity is $G = 1 - \sum p^2$, and the split with the largest decrease in weighted Gini is preferred. An illustrative first three levels are:

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ROOT: Credit Score <= 650
  ■■■ Yes (higher risk)
  ■ ■■■ Repayment History = Poor
  ■ ■ ■■■ Debt > 20,000 → Default
  ■ ■ ■■■ Debt <= 20,000 → Default/Monitor
  ■ ■■■ Repayment History = Good
  ■ ■■■ Loan Amount > 15,000 → Monitor
  ■ ■■■ Loan Amount <= 15,000 → No Default
  ■■■ No (lower risk)
  ■■■ Existing Debts > 30,000
  ■ ■■■ Income <= 40,000 → Monitor
  ■ ■■■ Income > 40,000 → No Default
  ■■■ Existing Debts <= 30,000
  ■■■ Repayment History = Poor → Monitor
  ■■■ Repayment History = Good → No Default
```

Illustrative root comparison: Credit Score Gini decrease = **0.118**, Repayment History = 0.101, Existing Debts = 0.073, Income = 0.061. Since Credit Score gives the largest impurity reduction, it is selected as the root in this worked example.

(b) Evaluation and false negatives

Metric	Illustrative result
Accuracy	89%
Precision	84%
Recall	81%

F1-Score	82.5%
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A **false negative** is a real defaulter predicted as non-defaulter. These are especially costly because the bank may approve a risky loan. To reduce false negatives, use class weights, SMOTE on training data, threshold adjustment, and optimize recall/F1 or a cost-sensitive objective. A slightly lower overall accuracy may be acceptable if default detection improves substantially.

(c) Post-pruning

Pre-pruning restricts tree growth using limits such as maximum depth or minimum samples per leaf. Post-pruning first grows a larger tree and then removes branches using cost-complexity pruning. Illustrative result: pre-pruned accuracy = **87.5%**, post-pruned accuracy = **89.0%**. The post-pruned tree is preferred because it is simpler than the full tree, reduces overfitting, and is easier to explain during loan-approval review.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression comparison

Logistic Regression estimates the probability of default using a sigmoid function: $P(\text{Default}) = 1/(1+e^{-z})$, where $z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$. It is suitable when a relatively simple, interpretable relationship between predictors and log-odds of default is adequate.

Model	Accuracy	AUC-ROC
Decision Tree (illustrative)	89%	0.91
Logistic Regression (illustrative)	91%	0.94

In this illustrative example, Logistic Regression has better accuracy and AUC-ROC. A statistical model is preferable when probability estimates, coefficient-level interpretation, stable behaviour, and a relatively linear relationship are important. The Decision Tree is preferable when nonlinear interactions and rule-based explanations are more useful.

(b) Feature importance

Decision Tree top 3: Credit Score, Repayment History, Existing Debts. **Logistic Regression top 3:** Credit Score, Repayment History, Loan Amount. They agree on the first two important predictors but differ on the third. This is reasonable because tree importance is based on impurity reduction, whereas Logistic Regression importance is inferred from standardized coefficient magnitude. Different models can therefore rank correlated predictors differently.

Task 4: Reinforcement Learning for Action Recommendation

(a) MDP formulation

Component	Definition
States	S1: Good repayment, S2: Watch, S3: Risky, S4: Severe risk
Actions	Approve, Reject, Restructure, Monitor
Reward	+10 for improved repayment; +5 for stable repayment; -10 for deterioration; -20 for severe loss
Transitions	Action changes the customer's next repayment state according to observed/estimated behaviour

The state represents current repayment condition, the action is the bank's intervention, the reward measures financial/repayment improvement, and transition dynamics describe how the customer's state changes after an action. The MDP allows the agent to optimize long-term outcomes rather than one immediate decision.

(b) Q-Learning – 5 episodes and updates

Q-learning update: $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$. Use $\alpha=0.5$, $\gamma=0.9$ and ϵ -greedy exploration. Example training episodes: E1 S3→Monitor→S2, E2 S2→Restructure→S1, E3 S3→Restructure→S2, E4 S4→Restructure→S3, E5 S2→Monitor→S1.

Iteration	s	a	r	max $Q(s',a')$	Old Q	New Q
1	S3	Monitor	+5	4	0	4.30
1	S2	Restructure	+10	6	0	7.70
1	S4	Restructure	-10	5	0	-2.75
2	S3	Monitor	+5	7	4.30	7.80

2	S2	Restructure	+10	8	7.70	8.45
2	S4	Restructure	-10	4	-2.75	-1.70

Example calculation for iteration 1, S2–Restructure: $Q_{\text{new}} = 0 + 0.5[10 + 0.9(6) - 0] = \mathbf{7.70}$. Convergence criterion: stop when the maximum absolute change in Q-values across an episode falls below **0.01** for several consecutive episodes, or when the maximum episode count is reached.

(c) Final policy and ethics

State	Best action
S1 – Good	Approve / Monitor
S2 – Watch	Restructure
S3 – Risky	Monitor / Restructure
S4 – Severe risk	Reject / Restructure

Ethical deployment requires fairness, bias testing, explainability, human oversight, privacy protection, and auditability. The agent should not use protected characteristics or hidden proxies unfairly. High-impact credit decisions should remain reviewable by authorized humans, and customers should have an appropriate explanation and appeal mechanism. The RL system should be validated in simulation and monitored after deployment.